

Exercise 4:

Exercise 4: JOINS & UNIONS

4.1.1 (1)

```
SELECT EmployeeID, Firstname, Lastname, Department, Salary,
       ProjectID, ProjectName, Budget, Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	Firstname	Lastname	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Dee	IT	70000	101	AI Development	100000	Complete
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	John	Doe	IT	70000	103	System Audit	30000	Pending
5	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

412 (2)

```
SELECT EmployeeID, FirstName, LastName, Department, Salary,
ProjectID, ProjectName, Budget, Status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	10000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	20000	Pending
3	Bob	Johnson	Finance	75000	104	Financial analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk management	30000	Pending

413 (3)

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
ProjectName, Budget, Status
FROM Projects AS A
RIGHT JOIN Employees AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	EmployeeName	Last Name	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	10000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	20000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk management	30000	Pending

4-14 (4)
 SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
 ProjectName, Budget, Status
 FROM Employees AS A
 FULL OUTER JOIN AS B
 ON A.EmployeeID = B.EmployeeID

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	10000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	30000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

4.2.1 (5)

```
SELECT DISTINCT (City) AS Location  
FROM Employees  
UNION  
SELECT Status  
FROM Projects;
```

Location	Status
New York	Completed
Los Angeles	Ongoing
Toronto	Ongoing
London	Completed
Sydney	Completed

4.2.2 (6)

```
SELECT DISTINCT (City) AS Location  
FROM Employees  
UNION ALL  
SELECT Status  
FROM Projects;
```

Location	Status
New York	Completed
Los Angeles	Ongoing
Toronto	Pending
Sydney	Ongoing
New York	Completed
New York	Pending

4.3.1 (7)

```
SELECT EmployeeID, FirstName, LastName, Department, Salary  
FROM Employees  
WHERE SALARY > 70000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

4.2.3
~~4.2.2~~ (8)

SELECT EmployeeID, First Name, Last Name, Department, Salary
FROM Employees
WHERE Department = 'IT' OR Department = 'Finance';

EmployeeID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

4.2.4 (9)

SELECT ProjectID, Project Name, Budget, Status
FROM Projects
~~WHERE Status = 'Ongoing' AND~~
WHERE Status ~~NOT~~ != Completed;

ProjectID	Project Name	Budget	Status
102	Employee Training AI Development	50000	Ongoing
103	Cybersecurity Security Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
105	Risk Management	80000	Pending

4.2.5(10)

SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000 AND status != Completed;

ProjectID	ProjectName	Budget	Status
101	At Developer		

ProjectID	ProjectName	Budget	Status
103	CyberSecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

4.2.6(11)

SELECT EmployeeID, FirstName, LastName, Department, Salary,
City
FROM Employees
WHERE City = 'New York' OR City = 'Toronto'
ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
1	John	Doe	IT	70000	New York
3					

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

4.2.7 (12)

```
SELECT EmployeeID, FirstName, LastName, Department, Salary  
FROM Employees  
ORDER BY Salary DESC  
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

4.3-1 (13)

```
SELECT Department, SUM(Salary) AS TotalSalary  
FROM Employees  
GROUP BY Department;
```

Department	Salary
IT	142000
HR	60000
Finance	155000
Sales	65000

4.3.2(14)

SELECT City, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AverageSalary > 65000;

City	AverageSalary
New York	75000
Los Angeles	60000
Toronto	75000
London	72000
Sydney	65000

4.3.3(15)

SELECT Department, COUNT(EmployeeID) AS EmployeeCount
FROM Employees
GROUP BY Department
HAVING EmployeeCount > 1;

Department	EmployeeCount
IT	2
Finance	2

4-3-4(16)

~~SELECT~~ ~~Project~~

4-3-4(16)

SELECT Status, Count(ProjectID) AS ProjectCount
FROM Projects
GROUP BY Status
HAVING ~~Project~~ Project > 1;

Status	ProjectCount
Completed	2
Pending Ongoing	2
Pending	2

~~4-3-4(16)~~
~~SELECT~~

4.3.4(17)

```
SELECT SELECT EmployeeID, FirstName, LastName, SUM(Budget) AS  
TotalProjectBudget  
FROM Employees AS A  
LEFT JOIN Projects AS B  
ON A.EmployeeID = B.EmployeeID  
GROUP BY EmployeeID  
HAVING TotalProjectBudget > 150000
```

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000